

Visual Chart Appendix — Tables A–P

This appendix pairs every required Table A–P from the audit report with a visual chart. Quantitative charts use the report values. Qualitative tables use explicit status, evidence-strength, coverage, action, or priority encodings; these encodings are visual summaries, not new scientific measurements.

Table A — Repository Snapshot

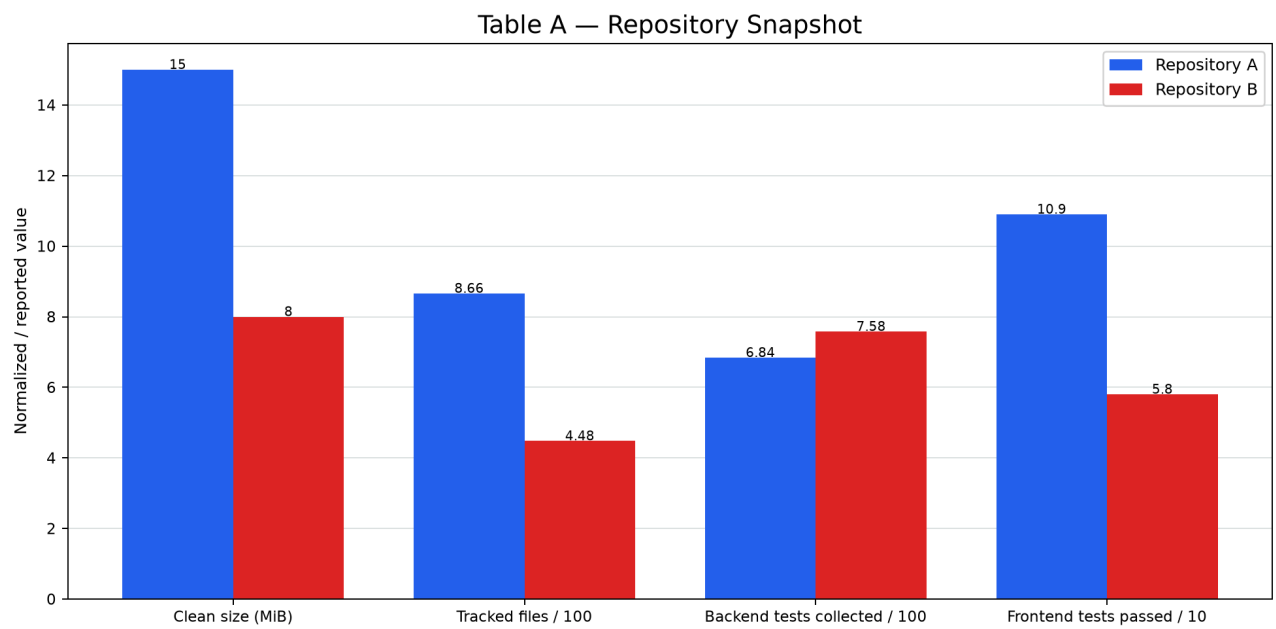


Table B — Docs vs Code vs Tests vs Runtime

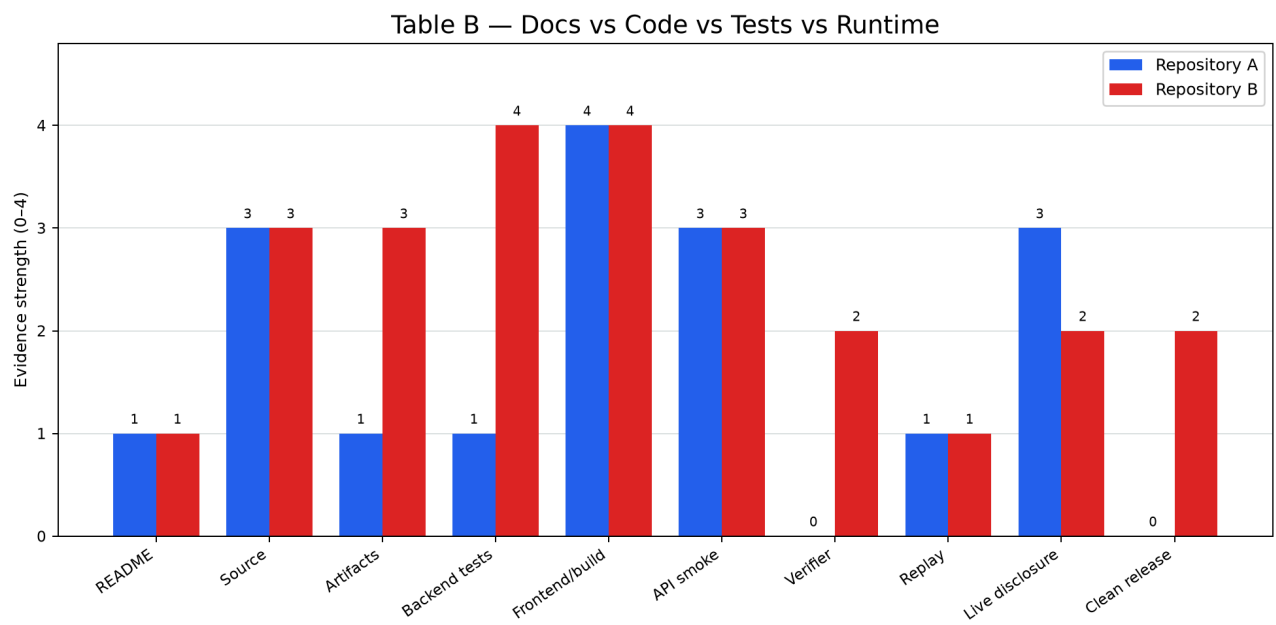


Table C — Major Capability Comparison

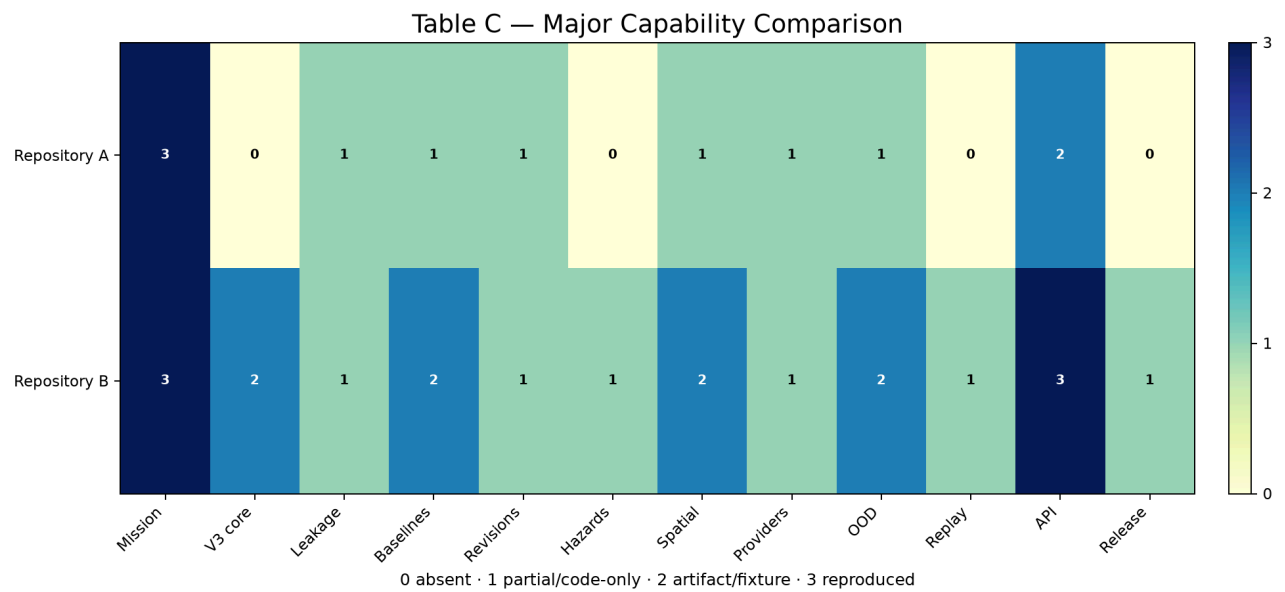


Table D — Main Docs Requirement Coverage

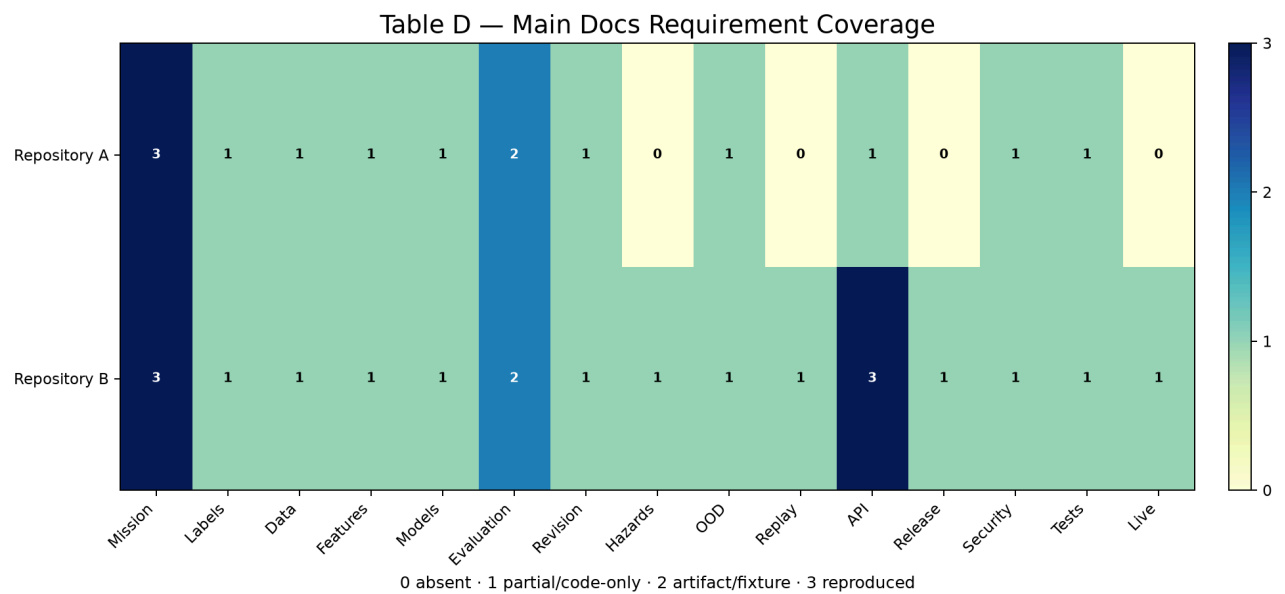


Table E — Hazard Specialist Evidence

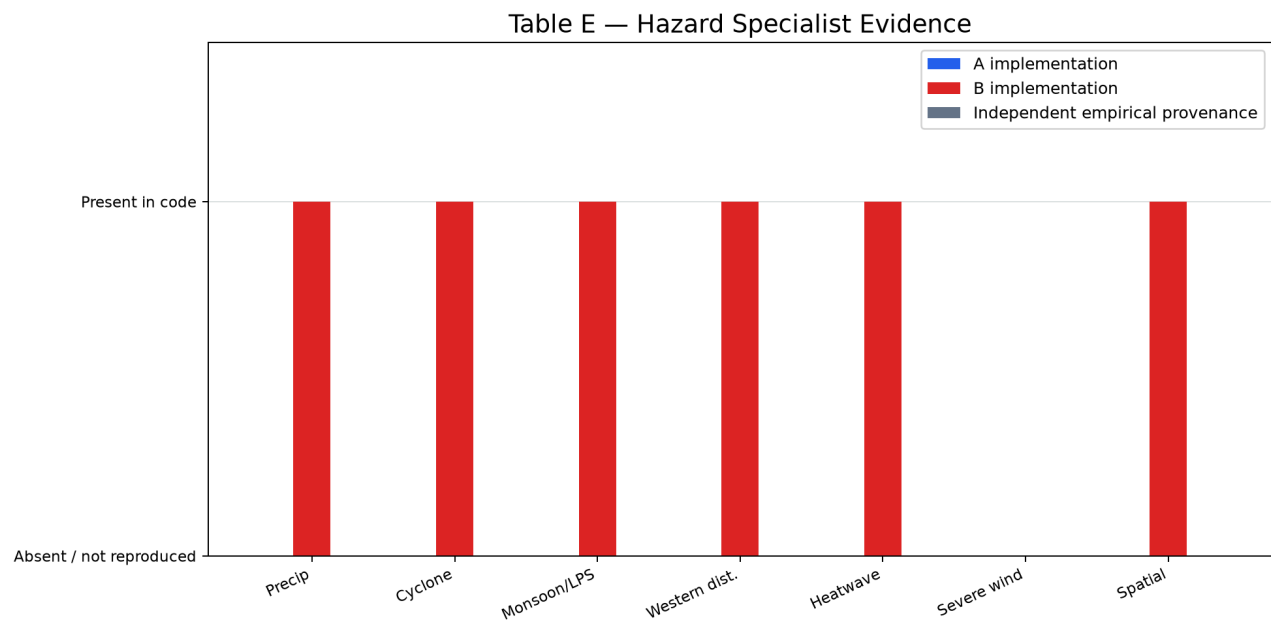


Table F — 500-Test Domain Coverage

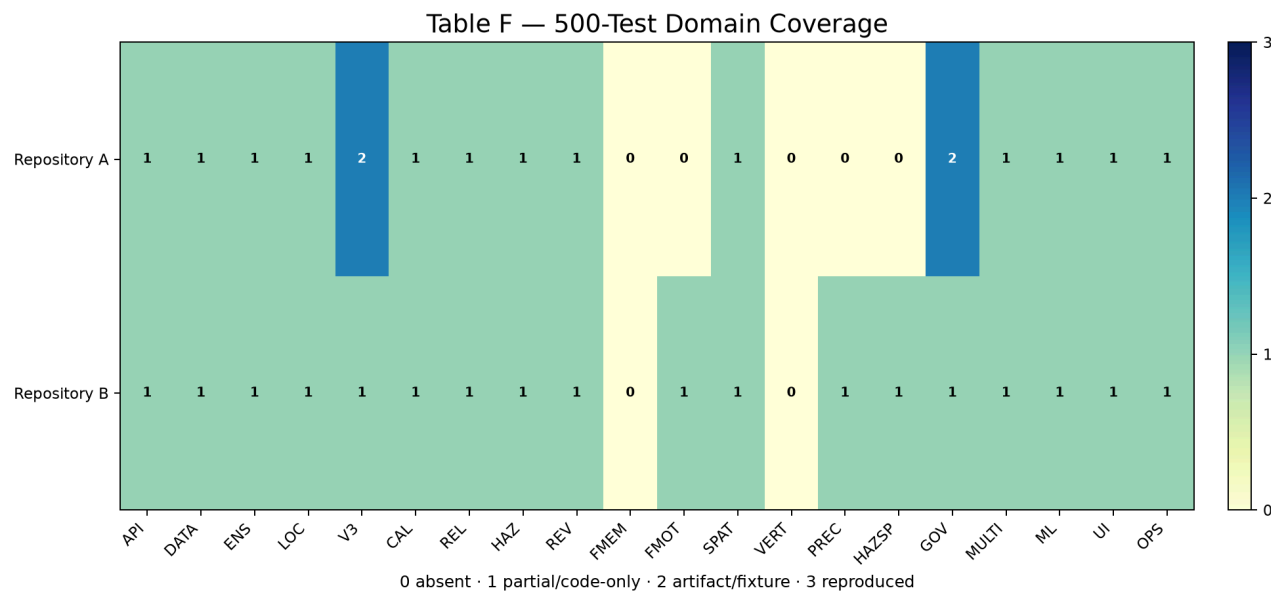


Table G — Scientific Claims: Claimed vs Proven

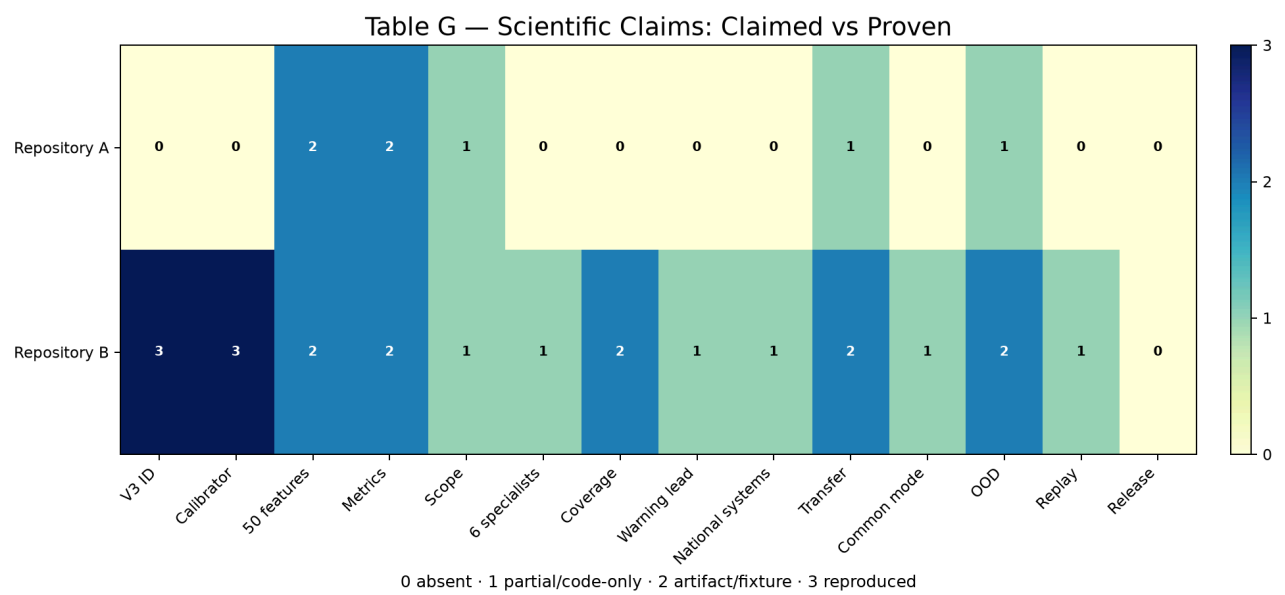


Table H — 0-100 Scoring

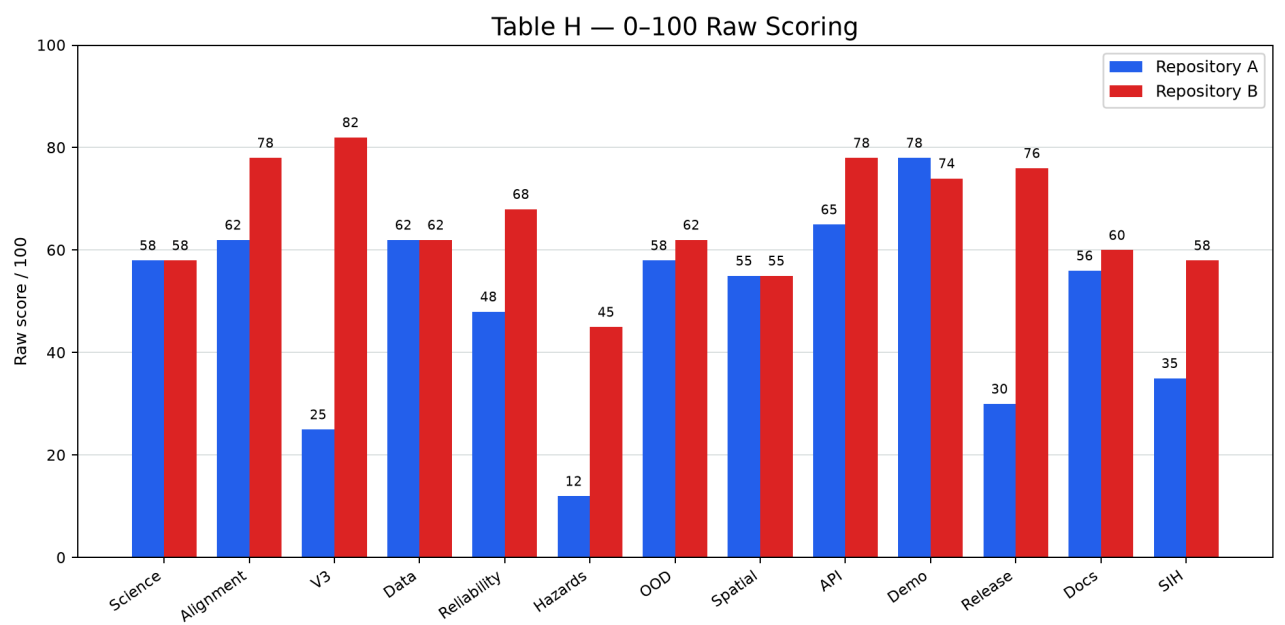


Table I — Weighted Total and Diagnostic Scorecard

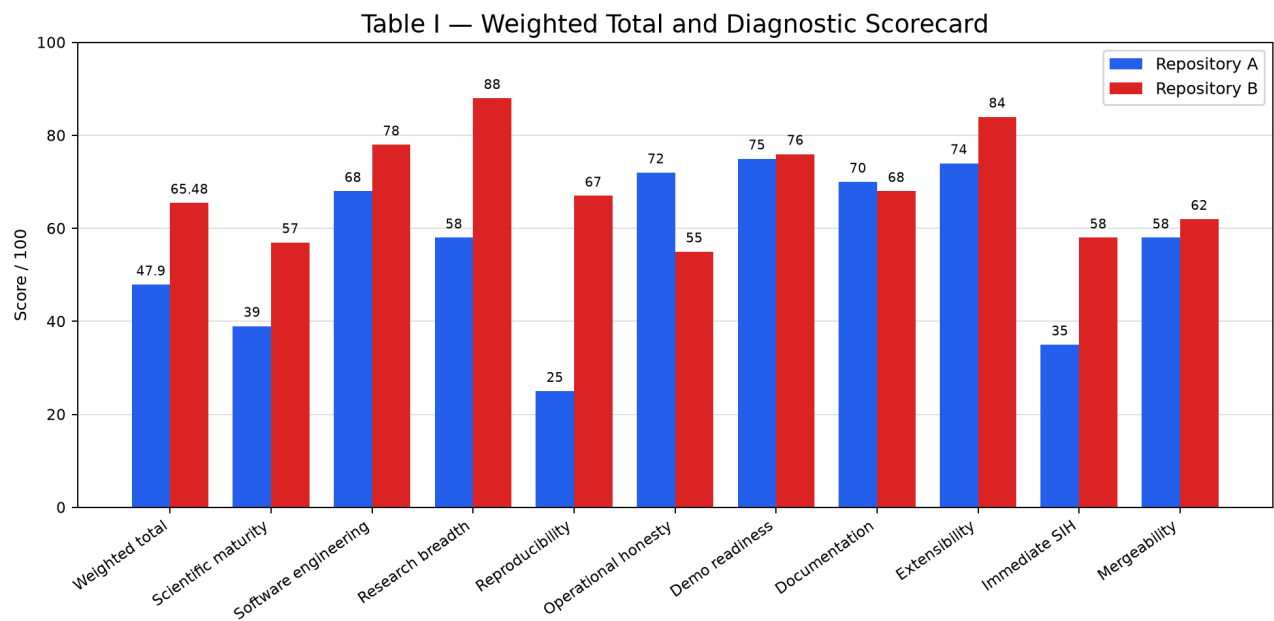


Table J — Selective Integration Actions

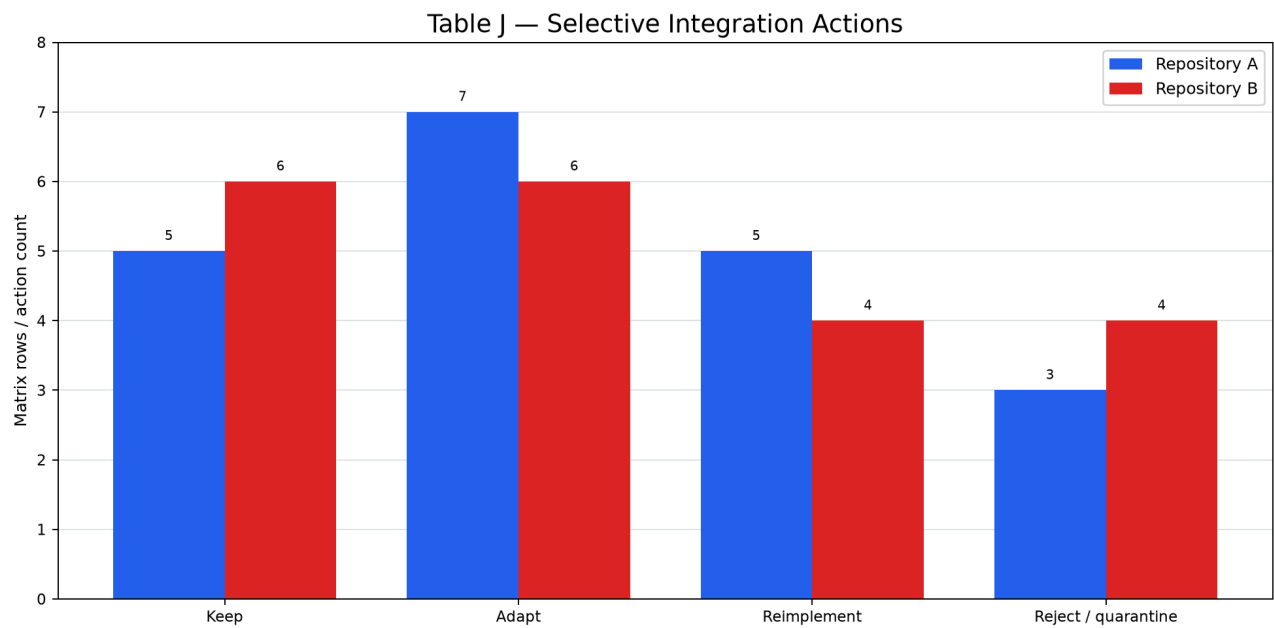


Table K — Mergeable Assets from Repository A

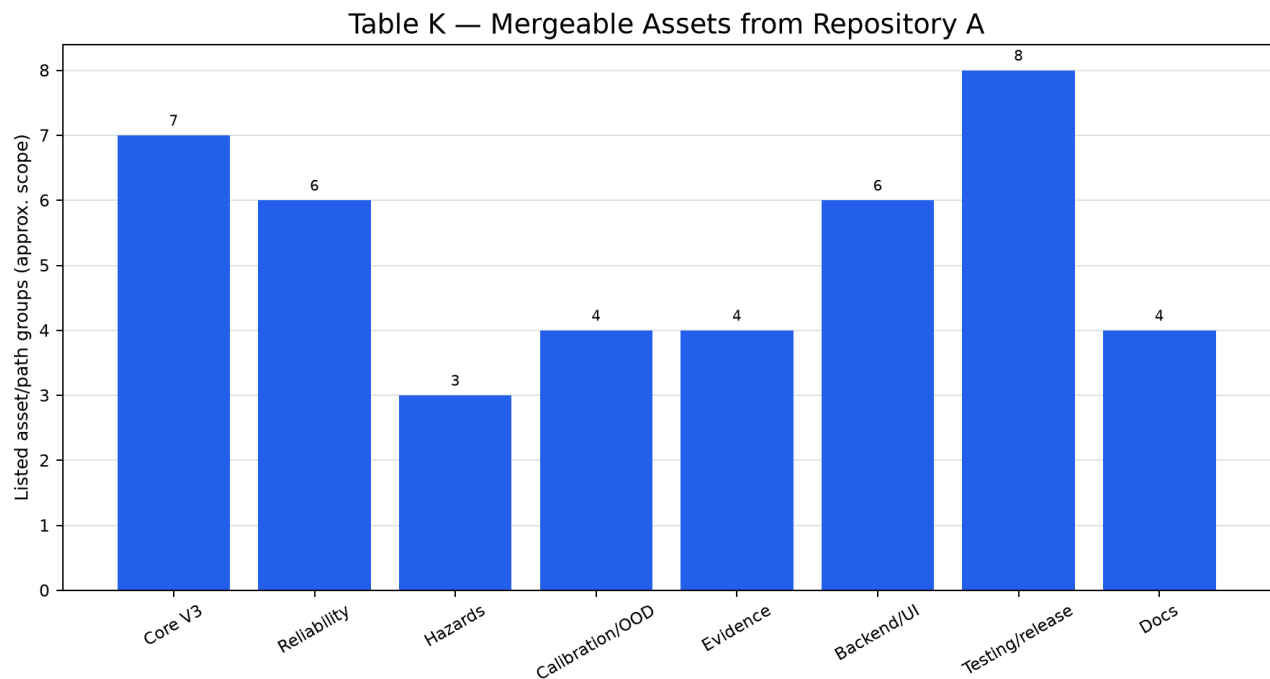


Table L — Mergeable Assets from Repository B

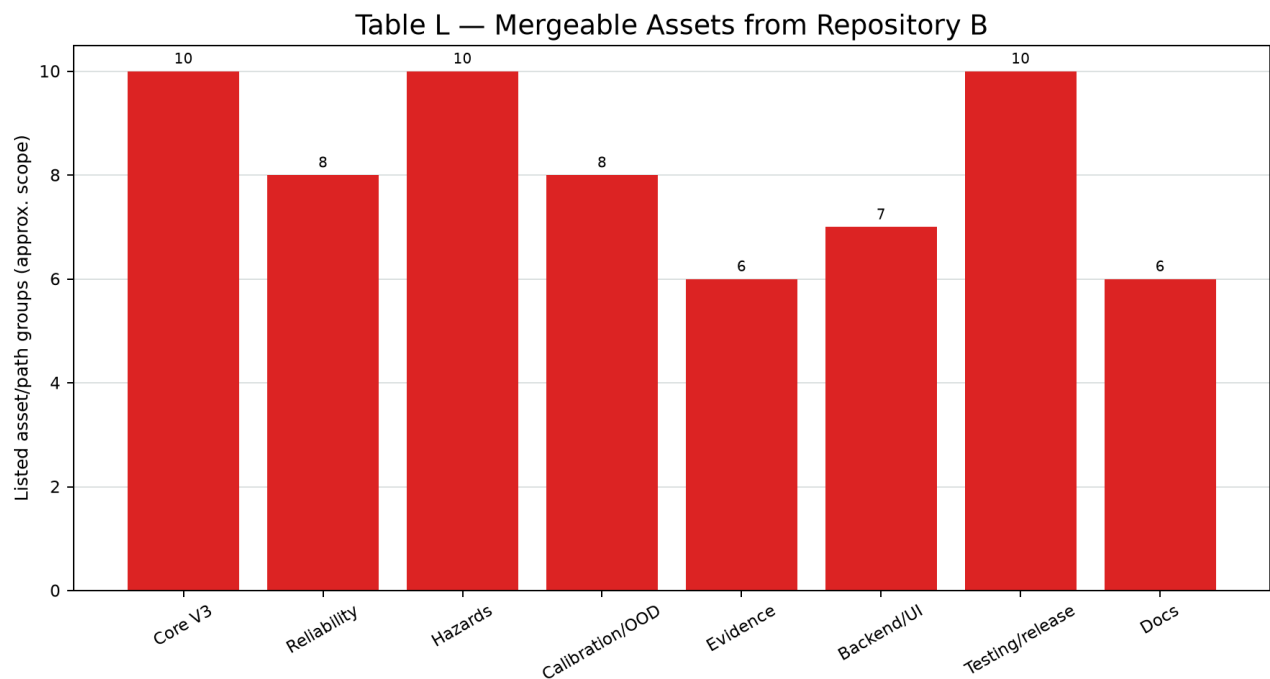


Table M — Merge Roadmap

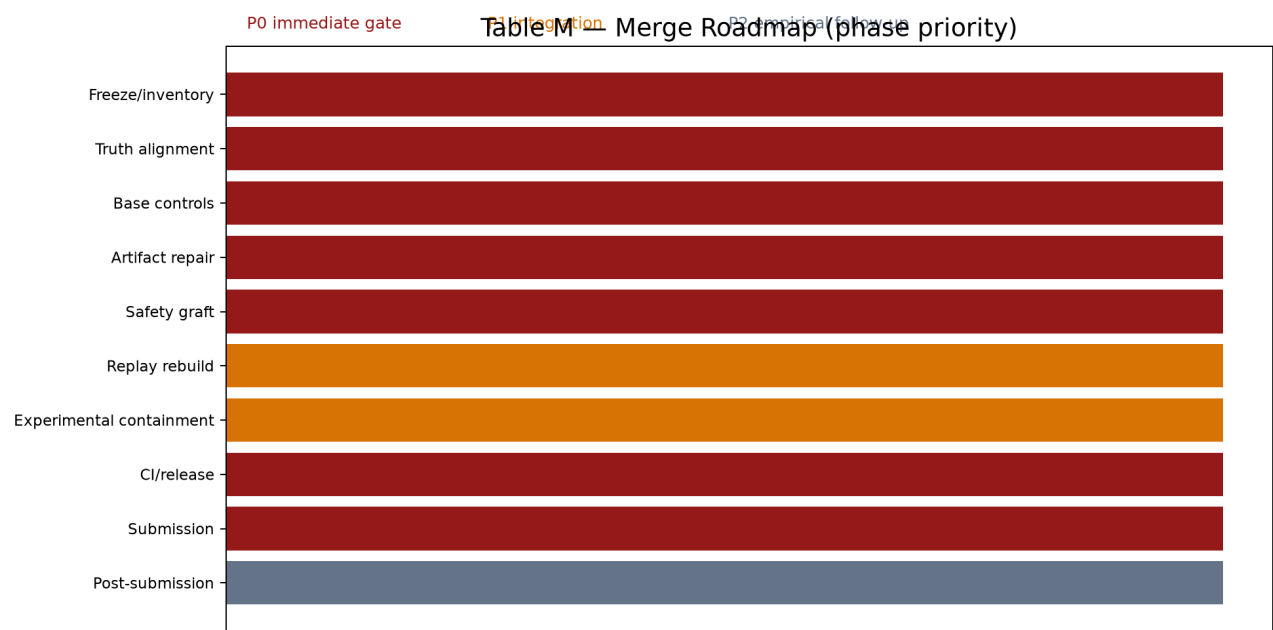


Table N — Validation Gates

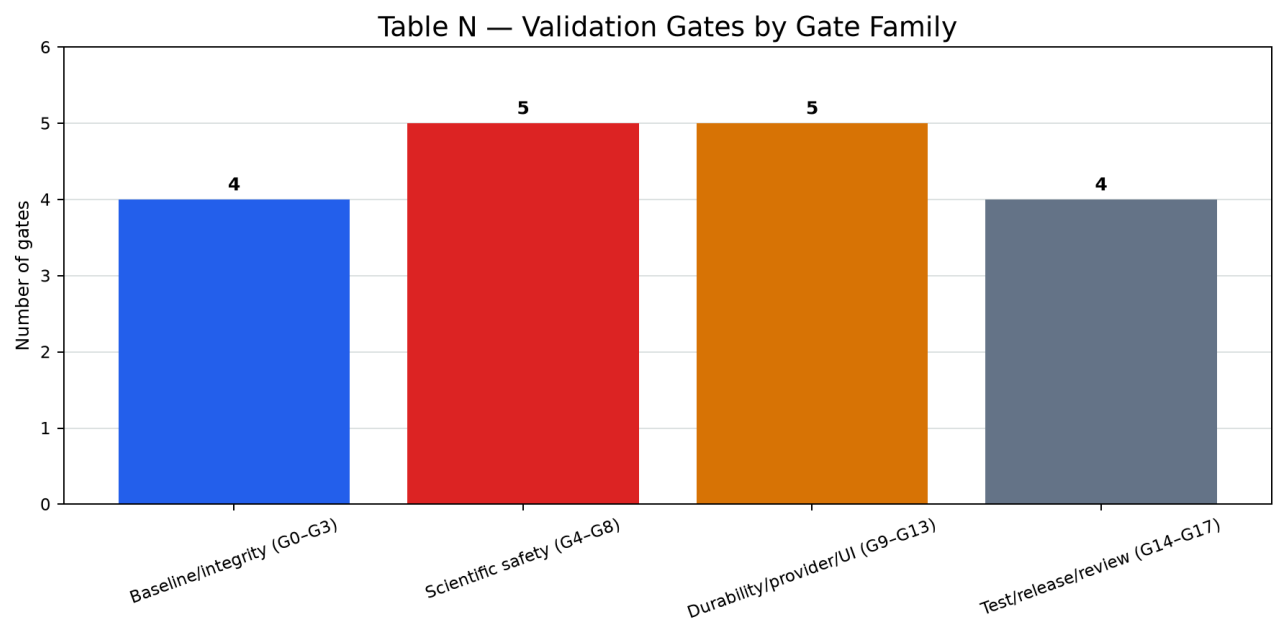


Table O — Merge Blockers

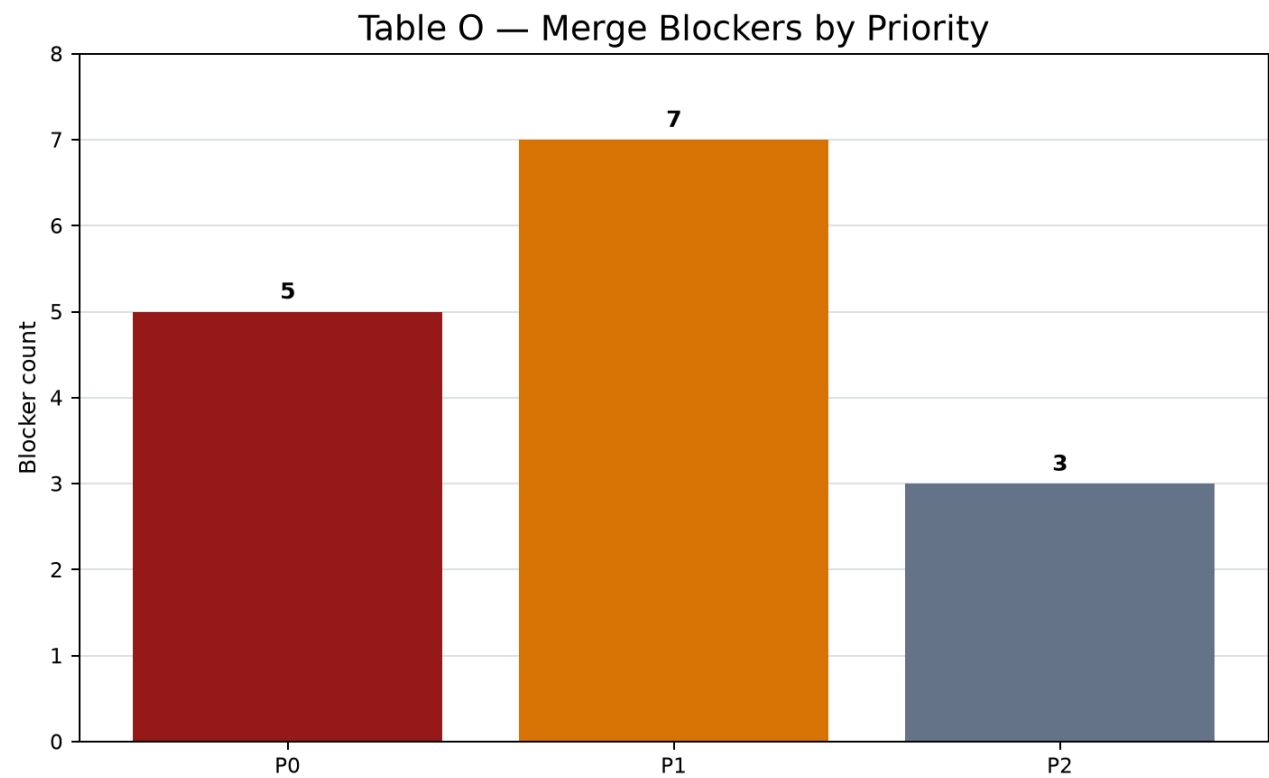
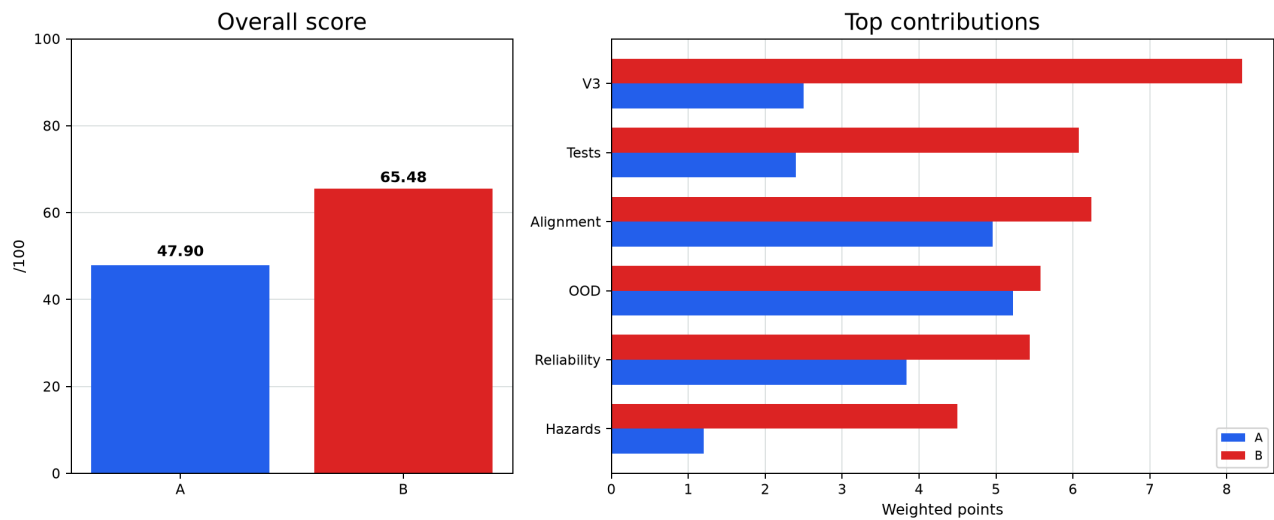


Table P — Graph / Scoreboard Summary



Source Audit Report

Final Evidence-Weighted SIH Round-2 Repository and Merge Report

Audit scope: Repository A and Repository B at the supplied, already-audited upstream default-branch heads

Audited Repository A SHA: b9f52d3eeec8676e06b1879f05b404605e2501be

Audited Repository B SHA: 82eded8194151e37fb9b3eecf273010dc62d7b29

Audit date: 2026-09-21

Reviewer role: Independent senior scientific-ML reviewer, software architect, release auditor, and integration planner

Author: Manus AI

Evidence order: reproduced execution > current source and committed artifacts > machine-readable outputs > code-tied technical documents > README/presentation claims > roadmap

Executive verdict

Repository B is the stronger base, the overall technical winner, the stronger current scientific foundation, the broader research architecture, and the better immediate candidate of the two. Neither repository is ready for unchanged SIH Round-2 submission. Repository A's default V3 path cannot load because its two committed `.joblib` files are Git LFS pointer text rather than the expected binaries. Exact on-disk hashes fail, replay fails, and serving correctly returns `MODEL_NOT_READY`. Repository B contains the expected, loadable V3 model and isotonic calibrator and reproduced **758 passing backend tests plus 58 passing frontend tests**, but its V3 feature-contract checksum fails. Its hazard specialists are deterministic formulas or baselines without specialist trained artifacts, and its certification, coverage, warning-lead, cross-system, common-mode, and replay claims exceed reproduced evidence.³

The immediate decision is `NO_REPOSITORY_READY_YET`. The lower-risk pre-submission path is to repair and harden Repository B in place, not merge application trees. The long-term decision is `KEEP_B_AS_BASE_IMPORT_FROM_A`, implemented only as a selective adaptation of Repository A's safe-failure, certification-boundary, live-versus-fixture disclosure, time-contract, and durable revision patterns. A monolithic merge would duplicate model-selection surfaces, APIs, frontends, data stores, tests, documentation, and deployment assumptions while creating no missing scientific evidence.

The final weighted scores remain $47.\frac{90}{100}$ for Repository A and $65.\frac{48}{100}$ for Repository B. Independent arithmetic verification gives weights totaling 100, Repository A 47.90 , Repository B 65.48 , and a B lead of 17.58 . No new observed evidence justifies changing the established category scores.³

Submission rule: Keep Repository B's frozen V3 as incumbent only after the feature-manifest chain passes. Treat specialist formulas, synthetic replay, cross-system metrics, coverage figures, warning-lead claims, and every `CERTIFIED` label as experimental or rejected until independently revalidated. Reject Repository A's pointer files as deployable binaries.

1. Audit method, status vocabulary, and scientific boundaries

The supplied documentation is a **target map**, not proof that either implementation satisfies it. Duplicate documents were not double-counted. The consolidated master specification, 120-file research study pack, hazard blueprint, 500-test specification, Docs-vs-Research-vs-Live comparison, Phase A–L material, repository evidence reports, and current source paths were interpreted through the same evidence hierarchy.^{1 5 7}

1.1 Status and claim classes

Term	Meaning in this report
VERIFIED	Directly reproduced at the audited SHA, within the stated scope.
IMPLEMENTED_NOT_INDEPENDENTLY_VALIDATED	Current code exists, but scientific or operational validity was not independently reproduced.
PARTIAL	Material implementation exists, but the required capability or evidence chain is incomplete.
PROTOTYPE_ONLY	Demonstration, heuristic, or deterministic research implementation; not validated production science.
DOCUMENTED_ONLY	Present in prose or roadmap, not established by current implementation evidence.
MISSING	Required implementation or evidence is absent.
N/A – uncovered/missing	The final repository has no named target; no equivalent is invented.
FUTURE_BY_DESIGN	Explicitly deferred by the governing specification.
CONTRADICTED	Higher-ranked executable or artifact evidence conflicts with the claim.
UNVERIFIED	Evidence is unavailable or insufficient to decide.

Scientific claims use `REPRODUCED` , `SUPPORTED_BY_ARTIFACT` , `SUPPORTED_BY_CODE_ONLY` , `SUPPORTED_BY_TEST_FIXTURE_ONLY` , `DOCUMENTATION_ONLY` , `CONTRADICTED` , or `UNVERIFIED` . A passing software test does not become empirical scientific validation. A JSON metric is artifact support, not a reproduced result. A `CERTIFIED` string is not certification.

1.2 Non-negotiable scientific boundaries

Veyra is a **forecast-reliability and bust-detection layer**, not a replacement weather forecaster. Hazard occurrence probability is not forecast-bust probability. Calibration, uncertainty, OOD, abstention, provider disagreement, ensemble spread, confidence, and certification are distinct concepts. Features must be available no later than issue time. A frozen incumbent may be replaced only after a challenger passes equivalent leakage, calibration, subgroup, uncertainty, OOD, and promotion gates.[1](#)

The prototype evidence boundary remains **public-proxy GEFS/Open-Meteo or WeatherBench 2 with ERA5/reference-analysis framing**. Neither repository proves paired NCMRWF/NEPS operational validation merely by naming NCMRWF, IMD, DWR, INSAT, ECMWF, or other systems. Rainfall reliability is not flood probability. Explanations identify predictive contributions or coded reason rules, not physical causality.[5](#)

2. Repository forensics and executable record

Table A — Repository Snapshot

Item	Repository A	Repository B
Neutral identifier	Repository A	Repository B
Upstream default branch	main	main
Audited/current upstream HEAD	b9f52d3eeec8676e06b1879f05b404605e2501be	82eded8194151e37fb9b3eecf273010dc62d7b29
HEAD timestamp	2026-09-21T09:44:50+05:30	2026-09-21T00:02:06+05:30
Visible HEAD subject	Merge of PR #55 for human-verification fixes	README scientific-workflow documentation update
Working tree at audit	Clean tracked tree	Clean tracked tree
Checkout/history depth	Shallow/grafted; one visible merge commit	Shallow/grafted; one visible commit
Branch/PR governance	One PR merge visible; protection/full history unverified	Direct push-to- main deployment visible; protection/PR history unverified
CI/workflow	No current .github/workflows	.github/workflows/deploy.yml builds/deploys frontend only
Initial clean-checkout size	15 MiB	8.0 MiB
Tracked files / tracked bytes	866 / 8,832,697	448 / 5,233,492
Top-level implementation	backend , frontend , models , data , scripts ; many historical/duplicate trees	backend , frontend , models , data , scripts , docs , .round2-roadmap
Backend	FastAPI under backend/app ; broad endpoints/services	FastAPI under backend/app ; versioned endpoints/services
Frontend	React/Vite; 9 executed test files	React/Vite; 4 executed test files
Model/artifact directories	models/v3 , models/day4 ; V3 binary paths contain pointer text	models/v3 , models/day4 , baseline logistic; V3 binaries load
Tests	Active pytest root backend/tests ; 684 collected	backend/tests ; complete run produced 758 passes
Deployment	vercel.json ; not independently deployed in audit	Pages workflow with external API URL; deployed environment unverified
Open release failure	V3 hash/load/replay/release failures	Feature-contract hash failure; verifier exits 1

File count, history depth, branding, README length, and number of future modules received no independent quality credit. Repository A's visible PR merge and Repository B's direct-main Pages deployment are governance signals only; neither proves protected-branch policy.

Table B — Docs vs Code vs Tests vs Runtime

Evidence layer	Repository A	Repository B	Disposition
README/roadmap	Root README retains stale 92-test language; historical trees coexist	README claims 816 tests and broad specialist certification/completion	Prose credited only where current code/execution agrees
Current source	Strict V3 adapter; issue-time contracts; certification/OOD policy; SQLite/WAL revision store; replay harness	Loadable V3 adapter; hazard, spatial, common-mode, evidence, drift, replay, and API/UI modules	Implementation credit only; formulas are not trained models
Committed artifacts	<code>models/v3/*.joblib</code> are LFS pointer text; manifests and feature file exist	Expected V3 model/calibrator binaries load; 50-feature file and many JSON manifests exist	A core identity contradicted; B model/calibrator reproduced, complete chain contradicted
Backend tests	684 collected; no complete passing run; targeted run had failures/errors	758 passed , 60 warnings, 343.24 s	B result is reproduced software evidence, not 758 scientific validations
Frontend tests/build	109 passed ; build passed; <code>React act(...)</code> and 698.05 kB chunk warnings	58 passed ; build passed; 655.97 kB chunk warning	Reproduced UI/build evidence
API smoke	<code>/v1/health</code> , <code>/v1/metrics</code> , <code>/v1/ood/policy</code> returned 200; health remained green while model unavailable	<code>/</code> , <code>/v1/health</code> , <code>/openapi.json</code> returned 200	Local startup only; A health is not model readiness
Artifact verification	Model/calibrator on-disk hashes fail; <code>joblib.Load</code> raises <code>KeyError: 118</code>	Model and calibrator pass; feature hash fails	Release blocker in both; decisive inference blocker in A
Replay	Independent harness structure, but current execution blocked by pointer files	Six-cycle script executes with source-declared synthetic progression and inconsistent Brier text	A current release claim contradicted; B fixture/simulation only
Live-vs-fixture	Open-Meteo live adapter; second provider explicitly fixture-only	Open-Meteo request code; synthetic fallback, deterministic risk map, and synthetic replay	No paired national-system validation in either
Clean clone/release	LFS payload unavailable; no CI; release manifest disk validation fails	Dependencies install and suites run; verifier/version/CI gaps remain	A contradicted for frozen release; B partial

Repository A's completed targeted backend run was **69 passed, 4 failed, 1 skipped, 9 errors**; a first-failure run was **28 passed, 1 failed**. Two bounded full runs did not finish, so no full-backend outcome is inferred. Repository B's backend run completed with **758 passed, zero failed/skipped/xfail**, and its frontend completed with **58 passed**. Repository A's frontend completed with **109 passed**. Test quantities are reported by command and are not treated as master-suite identity.³

3. Model and artifact audit

Repository A's paths `models/v3/lightgbm_v3_challenger.joblib` and `models/v3/probability_calibrator_v3.joblib` exist, but contain Git LFS pointer text. Their observed file hashes are `a6f97f085189f6c9f66dec162ba2052176b71d760656f4e6be1c95d51382d5d0` and `db4ba1d44201072322c0979abdd3a94f0497f9bc550558dcebd936d6ab832aa6`, not the expected model/calibrator hashes. Pointer OIDs matching expected hashes do not make the checkout files loadable artifacts.

`backend/app/builder2/v3_model_adapter.py` detects the mismatch and safely marks the model unavailable.
`backend/app/api/v1/endpoints/predict.py` does not silently substitute the legacy Day-4 model.[3](#)

Repository B's V3 model hash matches `00a8410746f4a0eecbf7e76aaa0565143fc948d0e06aea65e7bcc4ce28a1c660` ; the model loads as a LightGBM Booster. Its calibrator hash matches

`9f448606ce4338ded92f238a551b3a9d8e6d2cb5902e8bc687bce5f5850af531` ; it loads as `IsotonicRegression` , with a scikit-learn 1.9.0-to-1.9.1 warning. The 50-feature file exists, but its actual hash

`702ff4153fd95d8c9de3bbd01461d65fde0ef207099f7f3a8e7f5c8bac02031e` conflicts with declared

`265cffbbd157a2b8b8b46d3702438050980043b5ed3a6a646a7969cdb985335` . `scripts/verify_artifacts.py` therefore exits

[1.4](#)

Repository B also contains `models/day4/lightgbm_bust_model.joblib` , a Day-4 calibrator, and

`models/baseline_logistic_v1.joblib` . The V3 adapter uses threshold `0.060` , while a legacy Day-4 service uses `0.280` .

These values may both be legitimate in different model contexts, but release authority is ambiguous until a versioned route-to-model/threshold/fallback manifest defines which path serves each endpoint.

Table C — Major Capability Comparison

Capability	Repository A	Repository B	Evidence-weighted finding
Reliability-layer mission	VERIFIED as product contract	VERIFIED as product contract	Tie; no credit as a replacement forecaster
Public-proxy boundary	Explicit live/fixture disclosure	Provider names plus live and synthetic paths	A is more conservative; neither proves national-system validation
Frozen V3 core	CONTRADICTED : pointer files cannot load	PARTIAL : model/calibrator load, feature checksum fails	B clearly preferable after integrity repair
Feature order/calibration	Strict 50-feature checks; calibrator unavailable	50 entries; isotonic loads; chain mismatch	B preferable, not fully release-valid
Issue-time/leakage safety	UTC contract and anti-leakage tests; full lineage not regenerated	Split manifests, forbidden-feature tests; full lineage not regenerated	Similar PARTIAL evidence
Baseline ladder	V3/legacy assets, incumbent unavailable	Logistic, Day-4, V3 artifacts and scripts	B broader; evaluation still needs rerun
Revision intelligence	SQLite/WAL exact-target pattern; live single-cycle revision features zeroed	Revision features and Zarr store; durability/restart proof absent	A pattern preferable; both partial
Failure Memory/motifs	Revision/fingerprint modules; no live validated memory	Dedicated memory/motif modules and catalogue	B broader; neither scientifically established
Multi-horizon/recovery	Timeline/horizon code; no recovery validation	Horizon/recovery engines and API fields	B broader, unvalidated scientifically
Hazard specialists	Named empirical specialists absent	Six broad deterministic/formula paths	B architecture only; no certification credit
Spatial/compound/common-mode	Spatial endpoint/UI; compound/common-mode missing	Graph, compound, common-mode modules with fixed logic	B broader prototype
Provider/cross-system	Live primary plus explicit fixture second provider	Cross-system engine and evidence JSON	Neither proves paired transfer; A disclosure safer
OOD/abstention	OOD diagnostic-only; model failure safely abstains	Active hazard OOD/abstention with fixed thresholds	B broader; representative calibration missing
Drift/independent truth	Mostly planned	Monitor, ledger, independent-truth code	B implementation lead, evidence incomplete
Explanation/evidence graph	Deterministic fingerprints/reason codes	Explanation and evidence-graph modules	B broader; neither establishes causality
Replay/digital twin	Stronger harness concept, blocked at runtime	Executable synthetic replay	B demo lead; neither proves historical replay science
Backend/API	Broad but incomplete execution/readiness observability	Broad, local smoke, completed suite	B preferable
Frontend/demo	109 passes and build	58 passes, build, broader surfaces	A has deeper reproduced UI tests; B has broader integration

Capability	Repository A	Repository B	Evidence-weighted finding
Release/security/operations	No CI; release manifest invalid	Partial frontend CI; security tests; verifier invalid	B preferable but not ready

4. Document-to-code and Phase A–L coverage

Source abbreviations are: **S1** main Forecast-Bust Sentinel specification; **S2** 120-file research study pack; **S3** 500-Test Master Suite; **S4** hazard blueprint; **S5** Docs-vs-Research-vs-Live comparison; **MP** authoritative master prompt. Exact subsection-to-module mapping is marked uncertain where the evidence does not establish it.

Table D — Main Docs Requirement Coverage

Requirement/capability	Source	Repository A evidence / status	Repository B evidence / status
Mission, scope, SIH alignment	S1 §3; S4 §1	Reliability API/UI; VERIFIED as scope	Same; VERIFIED as scope
Bust definition/labels	S1 §8; S4 contracts	V3 manifests/label code; PARTIAL	data/target_manifest.json; targets; PARTIAL
Data sources/engineering	S1 data; S3 DATA	Historical/QC/live adapters; PARTIAL	Data/reference/hazard manifest service; PARTIAL
Feature engineering/order	S1 §9; MP model audit	Strict adapter and 50 names; binary parity unavailable; PARTIAL	Loadable V3 and 50 names; PARTIAL
Models/baseline ladder	S1 §10; S4 §5	Legacy/V3 paths; incumbent unavailable; PARTIAL	Logistic/Day-4/V3; PARTIAL
Evaluation science	S1 evaluation; S4 §5	Frozen metric manifests only; SUPPORTED_BY_ARTIFACT	V3/specialist JSON metrics; SUPPORTED_BY_ARTIFACT
Revision/historical intelligence	S4 §6; S3 REV	revision_store.py ; live features zeroed; PARTIAL	Zarr/revision features; restart PARTIAL
Failure Memory	S4 §6.1; S3 FMEM	Modules/history concepts; PARTIAL	failure_memory.py ; provided PARTIAL
Failure Motifs	S4 §6.2; S3 FMOT	Heuristic instability_fingerprint.py ; IMPLEMENTED_NOT_INDEPENDENTLY_VALIDATED	failure_motifs.py and c
Multi-horizon/time-to-bust/recovery	S4 §6.4; S3 HAZ	Timeline/horizon contracts; PARTIAL	Horizon/recovery tests/service IMPLEMENTED_NOT_INDEPENDENTLY_VALIDATED
Precipitation reliability	S4 §7; Phase D	No named specialist; N/A – uncovered/missing	Formula/baseline specialist; IMPLEMENTED_NOT_INDEPENDENTLY_VALIDATED
Cyclone reliability	S4 §8; Phase E	N/A – uncovered/missing	Deterministic track/intensity likelihood PROTOTYPE_ONLY
Monsoon/LPS reliability	S4 §9; Phase F	Docs/utilities only; N/A – uncovered/missing	Hand-coded target/regime likelihood IMPLEMENTED_NOT_INDEPENDENTLY_VALIDATED
Western disturbance	S4 §10; Phase G	N/A – uncovered/missing	Fixed decomposed probability PROTOTYPE_ONLY
Heatwave	S4 §11; Phase H	N/A – uncovered/missing	Explicit fixed logistic equation SUPPORTED_BY_CODE_ONLY

Requirement/capability	Source	Repository A evidence / status	Repository B evidence / status
Severe wind	S4 §12; Phase H	Generic wind variable only; N/A – uncovered/missing	Compound/heatwave-linked
Spatial reliability	S4 §§14–19; S3 SPAT	Endpoint/UI/tests; IMPLEMENTED_NOT_INDEPENDENTLY_VALIDATED	Graph engine/topology/tests;
Compound/common-mode	S4; Phase I	N/A – uncovered/missing	Fixed/copula/common-mode PROTOTYPE_ONLY
Multi-provider/cross-system	S1 E8; S3 MULTI; Phase K	Fixture second provider; PARTIAL	Engine/artifact; transfer DOCUMENTED_ONLY /artifact-
Drift/independent truth	S4; Phase J	OOD policy; truth future; PARTIAL	Monitor/ledger/audit module;
Explainability/evidence graph	S1; Phase L	Fingerprints/reason codes; code-only	Evidence graph/explainer; co
Digital twin/replay/counterfactual	S4; Phase L	Harness blocked; CONTRADICTED current execution	Synthetic replay; PROTOTYPE
API/backend	S3 API/OPS	Broad code, partial smoke; PARTIAL	Smoke and 758-test run; VE
Frontend/SIH demo	S3 UI	109 tests/build; VERIFIED locally	58 tests/build; VERIFIED lo
Reproducibility/release	S3 GOV/ML/OPS	LFS payload absent, no CI; CONTRADICTED	Verifier fails, CI partial; PART
Security/robustness/operations	S3 OPS	Middleware/configuration and selected tests; PARTIAL	Middleware/auth/RBAC/secu operationally
Test-suite coverage	S3 all domains	684 collect, full run incomplete; PARTIAL	816 passes, no exact 500-ID
Live/runtime checks	MP hardened checks	Process health masks model unavailable; readiness CONTRADICTED	Local startup; deployment/liv PARTIAL
Documentation drift audit	S1 §30; S5	Stale test count and duplicate histories; CONTRADICTED in places	Certification/replay drift; CON
Scientific claim audit	MP scientific audit	Safer wording; incumbent fails	Usable core; specialist claim:
Scoring/decision logic	MP rubric	Same weights/evidence caps	Same weights/evidence caps
Combine/integration safety	MP merge rules	Useful patterns; incompatible artifacts	Better base; conflicting labels

Phase A–L disposition. Phase A is CONTRADICTED in A and PARTIAL in B because neither has a clean V3 artifact chain. Phases B and C are partial in both, with B broader. Phases D–H are missing as named specialists in A and implemented only as deterministic or

formula-based prototypes in B; severe wind remains partial. Phase I is mostly absent in A and prototype/code-only in B. Phase J is diagnostic/future in A and implemented but unvalidated in B. Phase K lacks paired transfer and complete promotion evidence in both. Phase L is blocked in A and synthetic in B. This mapping does not equate roadmap text with execution.⁵

5. Hazard and 500-test evidence

Table E — Hazard Specialist Evidence

Specialist	Repository A	Repository B implementation	Empirical provenance
Precipitation	No named hazard-conditional trained specialist; N/A — uncovered/missing	<code>backend/app/builder2/precipitation_specialist.py</code> contains deterministic baseline/score logic	JSON metrics/coverage only; no specialist trained artifact, feature order, training data, or rerun
Tropical cyclone	N/A — uncovered/missing	<code>cyclone_specialist.py</code> contains deterministic track/intensity rules and constants	Held-out/certification assertions not reproduced
Monsoon/LPS	Docs/utilities only; N/A — uncovered/missing	<code>monsoon_specialist.py</code> contains hand-coded target/regime logic	Metrics are artifact claims no trained artifact
Western disturbance	N/A — uncovered/missing	<code>western_disturbance_specialist.py</code> contains decomposed fixed probabilities and thresholds	Reported cycle/bootstrap values not rerun
Heatwave	N/A — uncovered/missing	<code>heatwave_specialist.py</code> contains fixed logistic equations	Coefficient provenance absent
Severe/high wind	Generic V3 wind variable is not a specialist	No independent trained specialist; wind appears in compound/heatwave paths	No independent model/calibrator/evaluation package
Spatial reliability	Endpoint/UI/tests; no benchmark rerun	<code>spatial_reliability_engine.py</code> , graph/topology, inverse-distance/fixed values	Empirical covariance and held-out bounds not reproduced

No current evidence supports six independently certified empirical meteorological hazard specialists in either repository. Repository B has broader executable software, but software completeness and scientific validation are separate.

Table F — 500-Test Domain Coverage

Domain	Repository A	Repository B	Coverage finding
API	Broad route tests; full suite incomplete	Dedicated API/contract tests in passing suite	Both PARTIAL against named 25 IDs
DATA	Historical/QC test families	Data contracts/services/test families	Both PARTIAL ; raw lineage not regenerated
ENS	Ensemble-derived features/tests	Ensemble service/tests	Both PARTIAL
LOC	Location/certification tests	Location tests/catalogues	Both PARTIAL
V3	Dedicated tests; current artifact tests fail	Tests pass, external verifier fails feature hash	A CONTRADICTED ; B PARTIAL
CAL	Calibrator unavailable	Isotonic loads; calibration tests/metrics not empirically rerun	A PARTIAL ; B PARTIAL scientifically
REL	Schemas/test family	Multiple reliability tests	Both PARTIAL ; B broader
HAZ	Horizon/UI code; named specialists absent	Horizon/recovery/routing tests	Both PARTIAL
REV	SQLite store and revision tests	Features/storage; exact named mapping uncertain	Both PARTIAL
FMEM	Code; no exact dedicated 25-ID map	Services/fixtures; no exact 25-ID map	N/A — uncovered/missing for exact mapping
FMOT	Heuristic fingerprints; no exact domain map	Motif tests/catalogue	A uncovered; B PARTIAL
SPAT	One named test family	Graph/leakage/propagation tests	Both PARTIAL ; B broader
VERT	No dedicated vertical-domain map found	No dedicated vertical-domain map found	N/A — uncovered/missing in both
PREC	No named specialist tests	Precipitation contract/target tests	A uncovered; B PARTIAL
HAZSP	Named specialist suites absent	Cyclone/monsoon/WD/heatwave/compound tests	A uncovered; B PARTIAL ; no empirical upgrade
GOV	OOD/release/replay tests; release fails	OOD/drift/truth/security tests; artifact gate fails	A CONTRADICTED ; B PARTIAL
MULTI	Fixture-based disagreement tests	Multi-system tests and transfer artifact	Both PARTIAL ; no paired live science
ML	Feature/leakage code; complete run unavailable	Leakage/split manifests/baselines	Both PARTIAL ; training matrix not rebuilt
UI	109 passes and build	58 passes and build	Reproduced local subsets; no 25-ID certification
OPS	Middleware/deploy config; no CI; release failure	Security tests/partial CI; no backend gate	Both PARTIAL

The 500-test specification defines **20 domains × 25 tests**. Repository B's 816 completed tests are genuine software evidence, but test quantity is not identity. Neither repository has a one-to-one record for all 500 specified IDs with passed, failed, blocked, skipped, xfail, and **N/A — uncovered/missing** outcomes.²

6. Scientific claims: claimed versus proven

Table G — Scientific Claims: Claimed vs Proven

Major claim	Repository A	Repository B	Required dispos
Frozen V3 model identity	CONTRADICTED : pointer hash differs; load fails	REPRODUCED : expected hash and Booster load	Use B binary or chain passes
Frozen calibrator identity/type	CONTRADICTED on disk; manifest says isotonic	REPRODUCED : expected hash and IsotonicRegression ; version warning	Pin environmen
50-feature contract	SUPPORTED_BY_ARTIFACT /code; binary parity unavailable	Count REPRODUCED ; chain CONTRADICTED by checksum	Reconcile cano file/hash
V3 Brier/ROC/ECE	SUPPORTED_BY_ARTIFACT	SUPPORTED_BY_ARTIFACT	Reproduce from evaluation rows
25-station, 3-variable, ≤240 h scope	Code/artifact only; model unavailable	Code/artifact only; evaluation not rerun	Preserve as boi claim, not fresh certification
Operational >240 h	Code-only and labeled uncertified	Code-only; model authority ambiguous	Keep explicitly i
Six certified specialists	Missing/unverified	Software code-only; certification DOCUMENTATION_ONLY	Remove/relabel evidence packa
Specialist Brier/ECE/PR-AUC	UNVERIFIED	SUPPORTED_BY_ARTIFACT	Independent rei required
Conditional/conformal ≥90% coverage	UNVERIFIED	SUPPORTED_BY_ARTIFACT	Reject productio pending rerun
+24 h to +96 h warning	UNVERIFIED	SUPPORTED_BY_TEST_FIXTURE_ONLY	Synthetic progr cannot support
NCMRWF/NEPS/IMD/DWR/INSAT ingestion	UNVERIFIED	DOCUMENTATION_ONLY /code references	Require actual data/metadata/
Open-Meteo/GEFS path	Code-only; one upstream fetch occurred in test	Code-only; external live prediction not audit evidence	Label provider/live/ca on every respor
Cross-system transferability	Fixture-only second provider	Transfer JSON SUPPORTED_BY_ARTIFACT	Paired-system independent rei required
Common-mode/compound certification	Missing	Logic code-only; certification unverified	Experimental or
OOD safety	Diagnostic policy/test fixtures; not auto-abstaining	Crash behavior reproduced in tests; thresholds unvalidated	Distinguish diag active OOD
Revision durability	SQLite/WAL code-only; restart not rerun	Zarr/code-only; restart not demonstrated	Rebuild and tes target restart be
Historical replay independence	Current execution CONTRADICTED by pointers	SUPPORTED_BY_TEST_FIXTURE_ONLY ; cycles explicitly synthetic	Reimplement w immutable forec
Digital-twin scientific conclusion	UNVERIFIED	Fixture-only; Brier text conflicts	Reject scientific
Release integrity	CONTRADICTED	CONTRADICTED at feature checksum	No release clair verifier passes

Major claim	Repository A	Repository B	Required dispos
README test claim	Stale/ CONTRADICTED	758+58 reproduced	Keep counts co specific

7. Scores and diagnostic trade-offs

Table H — 0-100 Scoring

Weighted contribution equals $\text{raw} \times \text{weight} \div 100$. Science-heavy rows are capped where evidence is code-, fixture-, or document-only.

Category	Weight	A raw	A weighted	A confidence	B raw	B weighted	B confidence	Evidence reason
1. Scientific correctness, claim discipline & leakage safety	15	58	8.70	Medium	58	8.70	Medium	A is conservative but unavailable; B overstates specialist validation
2. SIH26079 docs/research alignment	8	62	4.96	Medium	78	6.24	Medium	B covers more phases; breadth is not certification
3. Core V3/baseline, calibration & artifact reproducibility	10	25	2.50	High	82	8.20	High/Medium	A cannot load binaries; B loads two artifacts but feature hash fails
4. Data pipeline, issue-time contracts, provenance & QC	7	62	4.34	Medium	62	4.34	Medium	Controls exist; raw lineage/national paired data absent
5. Revisions, Failure Memory, Motifs, multi-horizon/recovery	8	48	3.84	Medium	68	5.44	Medium/Low	A has durable design; B broader modules; neither proves live history
6. Hazard specialists and empirical validation	10	12	1.20	High	45	4.50	High	A lacks specialists; B specialists are deterministic/formula-based
7. Certification, OOD, abstention, drift & independent truth	9	58	5.22	Medium	62	5.58	Medium	Safety paths exist; representative validation absent
8. Spatial, ensemble, provider & cross-system intelligence	6	55	3.30	Medium	55	3.30	Medium/Low	A is fixture-honest; B broader; neither proves transfer
9. Backend/API architecture & robustness	6	65	3.90	Medium	78	4.68	High/Medium	B completed suite/smoke; production operations untested
10. Frontend/demo & scientific communication	5	78	3.90	High	74	3.70	High	Both build; A has 109 tests, B broader surfaces
11. Testing, reproducibility, replay & release engineering	8	30	2.40	High	76	6.08	High/Medium	A backend/replay fail; B suites pass but verifier/CI incomplete
12. Documentation accuracy, traceability & maintainability	4	56	2.24	Medium	60	2.40	Medium	A stale/duplicated; B certification/replay drift
13. SIH Round-2 submission readiness	4	35	1.40	High	58	2.32	Medium	A inference unavailable; B needs integrity/claim cleanup
Total	100	—	47.90	Medium	—	65.48	Medium	B leads by 17.58

Table I — Weighted Total

Result	Repository A	Repository B	Interpretation
Final weighted score	47. ⁹⁰ / ₁₀₀	65. ⁴⁸ / ₁₀₀	A: partial/mostly prototype; B: functional but incomplete or weakly evidenced
Scientific maturity	39	57	B has a loadable incumbent; neither reproduces specialist science
Software engineering maturity	68	78	B has complete backend execution and broader modularity
Research breadth	58	88	B implements more Phase A–L surfaces
Reproducibility	25	67	A lacks usable V3 payloads; B has checksum/version issues
Operational honesty	72	55	A is clearer about fixtures/scope; B overuses certification language
Demo/presentation readiness	75	76	Both build; B is broader but needs simulation labels
Documentation quality	70	68	A is detailed but stale; B is extensive but claim-drifted
Long-term extensibility	74	84	B's modules/schemas are the broader base
Immediate SIH submission readiness	35	58	B is closer; neither is ready unchanged
Mergeability/integration compatibility	58	62	Modular, but contracts, thresholds, stores, and labels conflict

8. Immediate base selection and split combine decision

Q1 — Which repository is the stronger base?

Repository B. It has the expected loadable V3 model and calibrator, a completed backend and frontend test record, broader current modules, a functioning local API, and a more suitable destination architecture. The selection is not an endorsement of its specialist science. Repository A remains useful as a source of operational safety patterns, but a base whose default incumbent cannot deserialize is not the stronger current base.

Immediate suitability as-is

Required choice: `NO_REPOSITORY_READY_YET`. Repository A cannot demonstrate its central scientific model. Repository B can demonstrate V3 inference components and broad software, but its own verifier exits 1, specialist certification exceeds provenance, synthetic replay can be mistaken for historical evidence, CI omits backend/scientific gates, and model-selection authority is not frozen. The correct short path is a bounded Repository B repair sprint, not unchanged submission or a pre-submission cross-repository merge.

Hypothetical combined-system outcome

A raw merge would first make the system **less reproducible and maintainable** by duplicating FastAPI applications, React frontends, model services, stores, test suites, docs, and deployment assumptions. It would not repair A's absent binary payloads, B's feature checksum, B's specialist provenance, or either repository's missing paired national-system validation.

A selective combination can improve the system if Repository B remains the sole active base. Keep B's loadable V3, broad API, test suite, and experimental specialist architecture. Adapt from A only the safe `MODEL_NOT_READY` behavior, explicit certification-scope policy, fixture-provider disclosure, UTC/time contract, release/replay invariants, and exact-target SQLite/WAL revision semantics. Rebuild the release manifest, provider registry, revision service, and real replay rather than copying conflicting implementations. Revalidate every serving, calibration, leakage, OOD, provider, revision, and UI contract. The combination is worth the risk **only after immediate B blockers are fixed and only as selective integration**.

Table J — Selective Integration Matrix

Capability	Best source	Exact source modules	Scientific status	Action	Dis
Frozen V3 binaries	B	<code>models/v3/lightgbm_v3_challenger.joblib</code> ; <code>probability_calibrator_v3.joblib</code>	Identity reproduced	Keep	M
V3/release manifest	Neither unchanged	B <code>models/v3/*manifest*.json</code> ; A <code>backend/app/core/release_manifest.*</code>	Both chains fail	Reimplement	M
Serving authority	B destination	B <code>v3_model_adapter.py</code> , model registry/services; A <code>predict.py</code> failure semantics	Implemented, authority ambiguous	Reimplement	M
Issue-time/UTC contract	A pattern + B manifests	A <code>backend/app/core/time_contract.py</code> ; B training/target manifests	Code/artifact only	Adapt	L
Safe missing-model behavior	A	A <code>v3_model_adapter.py</code> and prediction safety layer	Runtime failure reproduced	Adapt	L
Certification boundary	A	<code>backend/app/core/certification_policy.py</code> ; certification endpoint/tests/UI repairs	Scope code-only	Adapt	L
Hazard specialists	B	<code>backend/app/builder2/*specialist.py</code>	Deterministic prototypes	Keep isolated/Adapt	L
Severe-wind specialist	Neither	B compound/heatwave pathways only	Independent specialist absent	Reject current claim	L
Revision durability	A pattern	A <code>revision_store.py</code> ; B <code>backend/app/data/zarr_store.py</code>	Both partial	Reimplement	F
Failure Memory/motifs	B architecture	B <code>failure_memory.py</code> , <code>failure_motifs.py</code> ; A <code>instability_fingerprint.py</code>	Heuristic/code-only	Adapt	M
Spatial reliability	B	<code>spatial_reliability_engine.py</code> ; <code>data/spatial_network_topology.json</code>	Unvalidated implementation	Keep isolated	M
Provider/cross-system	Neither unchanged	A provider adapters/disagreement; B <code>cross_system_transfer_engine.py</code>	Fixture/artifact-only	Reimplement	M
OOD/abstention	B destination, A scope tests	A <code>ood_policy.py</code> ; B <code>safety/ood_*</code> , hazard modules	Software safety only	Adapt	M
Explanation/evidence graph	B	<code>evidence_graph_engine.py</code> , <code>explainer.py</code> , explanation service	Predictive explanation only	Keep/Adapt	L
Historical replay	Neither unchanged	A replay harness/golden matrix; B digital twin/script	A blocked; B synthetic	Reimplement	F
Backend/API	B	<code>backend/app</code> , versioned routes/services	Locally reproduced	Keep	M
Frontend	B destination; selected A behaviors	Both <code>frontend/src</code> trees	Builds/tests reproduced	Adapt behavior only	M

Capability	Best source	Exact source modules	Scientific status	Action	D ris
CI/release gates	Neither complete	B <code>.github/workflows/deploy.yml</code> ; A no workflow	Incomplete	Reimplement	M
500-test record	Neither	Existing tests plus S3 IDs	No exact map	Reimplement ledger	L

9. Exact collection plans

The collection lists preserve evidence without implying that collected material is production-ready. **Canonical** means the item should be the destination's source of truth after validation. **Informative** means it supports adaptation or audit but must not be copied as authority.

Table K — Mergeable Assets from Repository A

Family	Exact assets/modules to collect	Why needed	Role/action
A. Core baseline/frozen V3	models/v3/feature_names.json ; training_manifest.json ; v3_evaluation_manifest.json ; backend/app/builder2/v3_model_adapter.py ; v3_feature_pipeline.py ; backend/app/api/v1/endpoints/predict.py ; models/day4/*	Compare strict schema/order, safe load failure, serving selection, historical baseline	Informative; Reject A V3 .joblib p files as binar
B. Reliability intelligence	backend/app/core/time_contract.py ; revision_store.py ; backend/app/services/revision_service.py ; backend/app/builder2/instability_fingerprint.py ; backend/app/core/replay_harness.py ; golden_replay_matrix.py	UTC identity, SQLite/WAL exact-target history, replay invariants, heuristic fingerprints	Informative; Adapt/Reim
C. Hazard specialists	backend/app/services/spatial_service.py ; spatial endpoint/schema/tests; provider-disagreement service/tests	A has no named hazard specialists; collect only useful spatial/provider safety behavior	Informative; Mark special families N/A uncovered/
D. Calibration/OOD/abstention	backend/app/core/certification_policy.py ; ood_policy.py ; V3 failure-safety/calibration/certification tests	Bounded scope, safe model-unavailable response, diagnostic OOD semantics	Canonical p Adapt
E. Evidence/explanation	backend/app/builder2/instability_fingerprint.py ; Audits/VEYRA_* ; backend/app/core/release_manifest.*	Reason-code pattern, prior artifact/release analysis, declared release checks	Informative; Adapt/Rejec claims
F. Backend/frontend/demo	Provider/certification/revision/spatial endpoints and schemas; frontend/src/test ; frontend/src/components/ModelCatalog.tsx ; ModelEvaluationView.tsx ; vercel.json	Proven UI behavior, scope labels, trust-state tests, deployment comparison	Informative; behavior on
G. Testing/reproducibility/release	backend/tests/test_v3_* ; test_day34_time_contract_revision_store.py ; test_day35_independent_replay_release_manifest.py ; test_day37_provider_adapters.py ; test_day38_cross_provider_disagreement.py ; test_scientific_certification.py ; pytest.ini ; requirements.txt ; frontend locks	Migrate safety and disclosure assertions missing from B	Canonical t port; Adapt

Family	Exact assets/modules to collect	Why needed	Role/action
H. Documentation	BUILDER_1_BUILDER_2_INTEGRATION_CONTRACT.md ; BUILDER_2_HANDOFF.md ; docs/HORIZON_REQUEST_CONTRACT.md ; relevant Audits/* ; current code-tied README sections	Preserve contracts and lessons, identify historical/stale statements	Informative; Preserve wi

Repository A explicit rejects: do not collect its two V3 pointer files as deployable artifacts; do not import historical duplicate application trees (Builder-2 , Parinidhi , Frontend-Original , Overview) wholesale; do not reuse the unresolved base-commit release manifest as authoritative; do not count the fixture second provider as live.

Table L — Mergeable Assets from Repository B

Family	Exact assets/modules to collect	Why needed	Role/action
A. Core baseline/frozen V3	models/v3/lightgbm_v3_challenger.joblib ; probability_calibrator_v3.joblib ; feature_names.json ; artifact_manifest.json ; training_manifest.json ; v3_evaluation_manifest.json ; v3_comprehensive_evaluation.json ; models/day4/* ; models/baseline_logistic_v1* ; backend/app/builder2/v3_model_adapter.py	Preserve loadable incumbent, baselines, model/calibration metadata	Canonical after repair; Keep
B. Reliability intelligence	failure_memory.py ; failure_motifs.py ; recovery/horizon_services/tests ; backend/app/data/zarr_store.py ; backend/app/services/drift_monitor.py ; independent_truth_audit.py ; motif/ledger data	Broad shared reliability architecture	Canonical experimental base; Keep/Adapt
C. Hazard specialists	Six backend/app/builder2/*specialist.py paths ; spatial_reliability_engine.py ; compound_hazard_engine.py ; common_mode_detector.py ; cross_system_transfer_engine.py ; hazard target/event JSON ; data/spatial_network_topology.json ; specialist JSON manifests	Preserve current prototype surface and contracts	Experimental; Keep isolated
D. Calibration/OOD/abstention	builder2/calibrator.py ; conditional_calibration_engine.py ; data/hazard_calibration_registry.json ; backend/app/safety/ood_detector.py ; ood_enforcement.py ; data/counterfactual_crash_test_report.json ; prediction schemas	Consolidate active safety behavior	Canonical implementation candidate; Adapt
E. Evidence/explanation	evidence_graph_engine.py ; explainer.py ; services/explainability_service.py ; provenance schemas/endpoints ; data/cross_system_evidence.json ; replay outputs	Keep provenance and non-causal explanation interface	Canonical UI/API pattern; Reject unsupported conclusions
F. Backend/frontend/demo	backend/app ; versioned routes ; frontend/src ; build/test scripts ; .github/workflows/deploy.yml ; Open-Meteo and fallback services	Destination application and demo	Canonical base; Keep
G. Testing/reproducibility/release	backend/tests ; frontend tests/locks ; scripts/verify_artifacts.py ; replay_digital_twin.py ; evaluate_cross_system.py ; evaluate_spatial_propagation.py ; smoke scripts ; requirements.txt ; pyproject.toml ; pytest.ini	Preserve reproduced suites and build the required gates	Canonical after correction; Keep/Extend
H. Documentation	ARCHITECTURE.md ; REPRODUCIBILITY_PACKAGE.md ; docs/* ; .round2-roadmap/* ; round2-report/* ; current README sections	Base architecture and roadmap evidence	Informative until claim cleanup; Adapt

Repository B explicit rejects or quarantines: do not promote specialist JSON metrics, cross-system values, conformal/coverage figures, warning-lead numbers, common-mode claims, or digital-twin conclusions as reproduced science. Do not let `fallback_service.py` synthetic cycles or deterministic risk-map outputs appear as live observations. Do not retain conflicting feature hashes or multiple undocumented serving thresholds.

10. Actual phased integration roadmap

The roadmap starts only after the immediate Repository B repair decision. Each phase corresponds to observed work; none assumes unavailable national or specialist data.

Table M — Merge Roadmap

Phase	Goal	Input from A	Input from B	Dependencies	Outputs
0 — Freeze and inventory	Freeze both SHAs and evidence before edits	SHA, hashes, pointer evidence, manifests, test logs	SHA, hashes, manifests, test/replay logs	Read-only clones; artifact inventory tool	Signed inventory, pa
1 — Truth alignment	Make claims match current evidence	Fixture/scope language; stale-README findings	Specialist/replay/transfer claims and docs	Claim taxonomy; source-to-claim owner	Claim register; stale/
2 — Base selection and branch controls	Establish B as sole destination and required checks	Selected pattern list only	Current B application tree	Clean Phase 0; protected integration branch	Base decision recor
3 — Incumbent artifact repair	Make frozen V3 chain authoritative without changing model	A strict failure/order test patterns	B model, calibrator, feature file/manifests, verifier	Exact environment; canonical feature decision	One release manifes ID/threshold/fallback
4 — Selective safety grafting	Adapt only A patterns that reduce operational risk	Time contract, safe failure, certification boundary, fixture disclosure, revision semantics	B registries, schemas, services, UI	Phase 3 frozen incumbent; schema design	B-native safety mod
5 — Reliability-store and replay rebuild	Create durable history and honest historical replay	SQLite/WAL exact-target design; replay invariants	B Zarr/history modules; synthetic engine as fixture only	Immutable event inputs; truth-latency contract	Durable revision sen historical replay mod
6 — Experimental module containment	Keep B breadth without overclaiming	A certification/display boundaries	B specialists, spatial, compound, common-mode, transfer, evidence graph	Namespaced experimental registry; feature flags	Formula-baseline lat disabled production
7 — Test, CI, and release consolidation	Turn evidence checks into required release gates	Ported A safety/release/provider/UI tests	B 758+58 suites, verifier, workflow, locks	Stable merged schemas; exact test map	CI for backend/frontend/bu 500-ID ledger; rollba
8 — Submission readiness	Produce a truthful, demonstrable release	Conservative claim language and UI scope checks	Repaired B demo, API, manifests, evidence package	All P0 gates; bounded demo script	Tagged candidate; ju risk register
9 — Post-submission empirical program	Decide whether any specialist deserves promotion	No A specialist artifacts	B formula baselines and contracts	Real issue-time forecast/reference data and governance	Per-hazard challeng reject

The roadmap deliberately separates **software integration** from **scientific promotion**. A specialist can be integrated as an experimental formula baseline while still failing the evidence needed for operational certification.

11. Validation gates and merge blockers

Table N — Validation Gates

Gate	Required evidence	Acceptance criterion
G0 Clean baseline	Fresh B clone at tagged commit and frozen A evidence bundle	Clean tree; dependencies from locked manifests; inventories match
G1 Artifact integrity	One manifest covering model, calibrator, feature order, environment, ID, threshold, fallback	Every SHA/type matches; <code>scripts/verify_artifacts.py</code> exits 0
G2 Incumbent parity	Golden inputs before/after integration	Identical calibrated V3 outputs within declared tolerance; no silent fallback
G3 Model authority	Route-to-model/threshold/fallback map	Exactly one authoritative model path per route; V3 remains incumbent
G4 Issue-time leakage	Raw-to-feature lineage and availability timestamps	No future observation, reanalysis, error, label, verifying imagery, or post-valid-time data
G5 Target semantics	Versioned generic/hazard contracts	Hazard occurrence never substitutes for bust; continuous error separate
G6 Calibration	Frozen OOT inputs/outputs	Brier, BSS, ECE, slope/intercept, reliability data, intervals reproduced
G7 OOD/abstention	Held-out shifts plus crash inputs	Risk-coverage reported; unsafe inputs abstain; diagnostic/active OOD distinct
G8 Specialist promotion	Per-hazard artifact/data/split/seed/bootstrap package	No <code>CERTIFIED</code> /production status until independent reproduction
G9 Revision durability	Multiple issue cycles, restart, truth sealing, exact-target lookup	Records reload with correct UTC/provider identity; no fabricated history
G10 Provider/cross-system	Aligned paired systems, metadata, independent truth	Live/fixture explicit; transfer metrics independently rerun
G11 Replay/digital twin	Immutable forecasts and independent truth	Historical mode contains no synthetic progression; metrics internally consistent
G12 API/schema compatibility	OpenAPI diff and consumer tests	No semantic drift; probability/uncertainty fields remain separate
G13 Frontend communication	Browser E2E for ready, abstain, OOD, live, cached, fixture, synthetic, unavailable	Correct scope/provenance banners; no false certification
G14 Test migration	Legacy-to-destination ledger and exact master-suite ID map	Critical tests pass; skipped/xfail/N/A justified; no generic substitution
G15 Security/operations	Required CI, secret/dependency scan, concurrency/rate/recovery tests	CI gates deployment; multi-worker limitations handled; rollback smoke passes
G16 Release governance	Required checks, signed/tagged release, rollback record	Failed science gate cannot deploy; prior version recoverable
G17 Independent review	Reviewer reruns hashes, tests, replay, build, smoke, claim register	Submission statements match evidence classes exactly

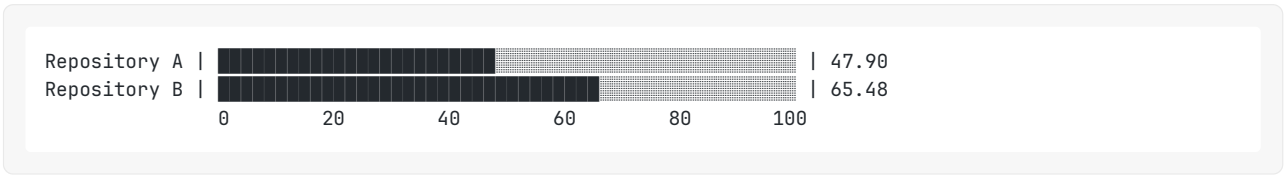
Table O — Merge Blockers

Priority	Blocker	Evidence	Required resolution
P0	A V3 binaries absent	Pointer text, wrong disk hashes, deserialization failure	Restore only for A archival verification; never copy pointers into B
P0	B V3 feature checksum fails	Actual 702ff415 ... ; declared 265cffbb ... ; verifier exit 1	Decide authoritative file/hash; regenerate manifest; clean-clone pass
P0	Specialist formulas lack empirical provenance	No specialist model/calibrator/training package	Relabel as formulas/prototypes or supply full evidence and rerun
P0	Certification/coverage claims exceed evidence	JSON/docs only; no independent result	Remove/restrict claims until G6/G8 passes
P0	Serving authority is ambiguous	V3 threshold 0.060 , legacy Day-4 0.280 , multiple services	One versioned route/model/threshold/fallback manifest
P1	Synthetic replay may look historical	Source declares synthetic progression; Brier text conflicts	Separate modes; visible labels; immutable historical replay
P1	CI lacks scientific/backend gates	B deploy workflow builds frontend only; A has no workflow	Required tests, verifier, security, replay, build, smoke before deploy
P1	Live/fixture/provider boundaries can be lost	A second provider is fixture; B has fallbacks/simulation	Provider provenance field and UI badge on every response
P1	Revision stores conflict	A SQLite/WAL vs B Zarr; key/sealing semantics differ	New canonical contract and restart migration tests
P1	API/probability semantics may drift	Duplicate schemas and hazard/bust/OOD fields	Contract-first migration and semantic assertions
P1	No exact 500-test map	Repository tests are not 500 named-ID certification	Produce ID-level outcome ledger
P2	Serialized-version warning	B calibrator created under sklearn 1.9.0, loaded under 1.9.1	Pin or rebuild under recorded compatible environment
P2	Frontend bundle warnings	A 698.05 kB; B 655.97 kB	Code-split or accept through measured performance gate
P2	Governance evidence limited	Shallow histories; no verified protection/rollback	Required checks, tags, provenance, rollback record

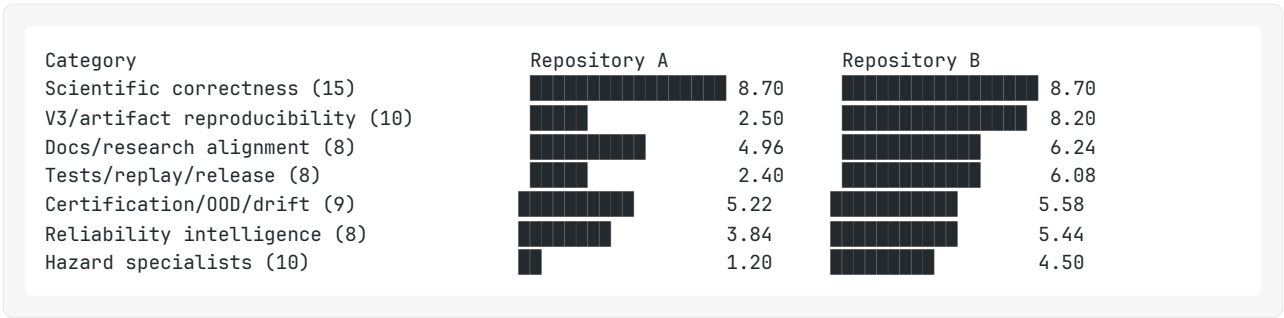
12. Scoreboard and readiness visualization

Table P — Graph / Scoreboard Summary

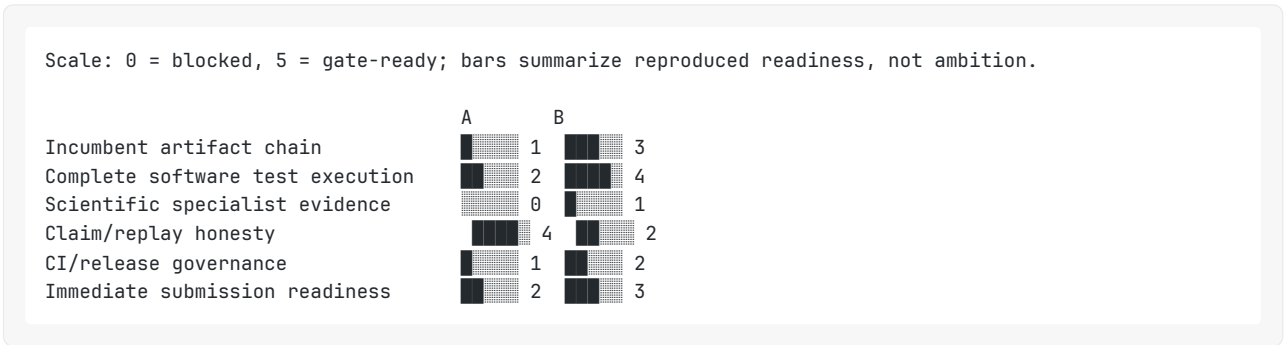
Overall weighted score



Top weighted contributions



Blockers/readiness



Compact conclusion: Repository B leads because its V3 model/calibrator load and its complete software suites execute. That lead does not validate its specialists. Both remain blocked from unchanged submission.

13. Symmetry check and required decisions

Identical scores, evidence caps, leakage rules, and scientific semantics were applied. B's deterministic specialists earned implementation credit but no trained-model or certification credit; the same cap would apply in A. Both failed release chains are marked contradicted, but A is penalized more because inference cannot deserialize, whereas B's model/calibrator work and only the feature-manifest chain fails. A receives higher frontend-test and operational-honesty credit even though B wins overall. B's 816 passes are not converted into 816 scientific validations. A's uncompleted backend run is not assigned an inferred pass count.

If labels were swapped but evidence stayed attached to the codebases, the codebase with loadable expected V3 binaries, 758 backend passes, 58 frontend passes, and broader modules would still win. The codebase with pointer-only V3 files, failing targeted release/replay checks, and no completed backend run would still lose. The conclusion does not depend on identity, branding, file count, README polish, prior human verification, or architectural ambition.

Required decision	Result	Basis
Overall Technical Winner	Repository B	65.48 vs 47.90 ; loadable incumbent, completed suites, broader implementation
Stronger Scientific Foundation	Repository B	Reproduced model/calibrator identity; A incumbent cannot load; specialist award excluded
Broader Research/Feature Architecture	Repository B	Wider Phase A–L code surface and schemas
Better Immediate SIH Candidate	Repository B	Closer after bounded repairs; not ready unchanged
Immediate Round-2 decision	NO_REPOSITORY_READY_YET	A core unavailable; B integrity and claim gates fail
Long-term decision	KEEP_B_AS_BASE_IMPORT_FROM_A	B is usable destination; A contributes selected safety patterns
Merge decision	SELECTIVE_MERGE_ONLY	Broad merge raises regression burden without adding evidence

14. Answers to Q1–Q5

Q1 — Stronger base: Repository B, for the loadable expected V3 artifacts, completed test execution, local runtime evidence, and broader modular architecture. This is an evidence decision, not an architecture or branding preference.

Q2 — What to merge and not merge: Keep B's V3 binaries, destination API/frontend, tests, and experimental module interfaces. Adapt A's model-not-ready handling, certification scope, live/fixture disclosure, UTC contract, replay invariants, and revision-store semantics. Reimplement the release manifest, route/model registry, provider registry, revision service, historical replay, and CI gates. Reject A's pointer artifacts and wholesale duplicate trees. Reject promotion of B specialist formulas, synthetic replay conclusions, and unsupported certification/transfer/coverage claims.

Q3 — Actual roadmap: Execute Phases 0–8 in Table M for a submission-quality selective integration, with Phase 9 reserved for post-submission empirical specialist work. Stop at any failure condition. Do not begin safety grafting until B's incumbent chain passes G1–G3.

Q4 — Exact materials to collect: Tables K and L provide the A–H collection plans with paths, canonical/informative status, action, and validation. The minimum A package is safety/time/certification/provider/revision/replay code plus its tests and code-tied contracts. The minimum B package is the V3/baseline artifact family, destination app, tests, specialist modules as experimental sources, manifests, scripts, locks, and architecture/reproducibility docs.

Q5 — Final target: A single Repository B-derived modular monolith with one authoritative V3 model registry and release manifest; strict issue-time and UTC/provider identity; safe model-unavailable behavior; durable exact-target revision history; explicitly separated live, cached, fixture, fallback, and synthetic states; experimental specialists isolated behind promotion gates; non-causal explanations; immutable real historical replay; mapped 500-test evidence; required CI, tagged rollback, and a claim register. No module may call itself certified without reproduced empirical evidence.

References

AUDITED_REPO_A_SHA: b9f52d3ecec8676e06b1879f05b404605e2501be
AUDITED_REPO_B_SHA: 82eded8194151e37fb9b3eecf273010dc62d7b29

STRONGER_BASE_REPO: Repository B
MERGE_DECISION:
- `SELECTIVE\_MERGE\_ONLY`

REPO_A_FINAL_WEIGHTED_SCORE: 47.90/100
REPO_B_FINAL_WEIGHTED_SCORE: 65.48/100

WHAT_TO_COLLECT_FROM_REPO_A:
- `backend/app/core/time\_contract.py`, `revision\_store.py`, `certification\_policy.py`, `ood\_policy.py`, `i
- `backend/app/builder2/v3\_model\_adapter.py`, `v3\_feature\_pipeline.py`, and `instability\_fingerprint.py`;
- `backend/app/adapters/openmeteo\_adapter.py`, `fixture\_second\_provider\_adapter.py`, and provider-disagree
- V3 failure-safety, certification, UTC/revision, provider-disclosure, replay/release, and frontend scope
- Current integration contracts and code-tied audit documents; reject stale claims and duplicate applicat
- Do not collect Repository A's V3 `.joblib` pointer files as deployable binaries

WHAT_TO_COLLECT_FROM_REPO_B:
- `models/v3/lightgbm\_v3\_challenger.joblib`, `probability\_calibrator\_v3.joblib`, `feature\_names.json`, mai
- `backend/app` as the destination application, including the V3 adapter, registry/services, API, schemas
- Hazard, spatial, compound, common-mode, cross-system, drift, independent-truth, explanation, and digit
- `backend/tests`, frontend tests/build configuration, `scripts/verify\_artifacts.py`, smoke/evaluation sc
- `ARCHITECTURE.md`, `REPRODUCIBILITY\_PACKAGE.md`, current roadmap/report files, and claim-audit inputs a

TOP_MERGE_PRIORITIES:
1. Repair Repository B's authoritative feature-contract hash and make the complete V3 artifact verifier e
2. Freeze one route-to-model, model-ID, threshold, calibrator, feature-order, and fallback manifest while
3. Adapt Repository A's safe model-unavailable, certification-boundary, UTC/provider-identity, and live-v
4. Rebuild durable revision history and immutable historical replay with independent truth; keep synthe
5. Gate deployment on mapped backend/frontend tests, artifact integrity, leakage checks, replay parity, s

TOP_MERGE_BLOCKERS:
1. Repository A's V3 files are Git LFS pointer text, not loadable binaries; its release and replay checks
2. Repository B's feature checksum disagrees with its manifests, so the complete frozen-core artifact cha
3. Repository B's specialists use deterministic formulas/fixed coefficients without trained artifacts or
4. Synthetic replay, cross-system values, coverage figures, warning-lead claims, common-mode claims, and
5. Model thresholds, API schemas, OOD semantics, provider identities, revision stores, dependency pins, C

HYPOTHETICAL_MERGED_SYSTEM_OUTCOME:
- A raw merge would initially weaken reproducibility and maintainability by duplicating model services, Al
- The fundamental benefit of a selective integration is Repository B's loadable V3 and broad implementati
- The fundamental risks are feature/target drift, future leakage, probability-semantic conflation, calibr
- Major conflicts include feature/release manifests, V3 versus legacy serving thresholds, diagnostic vers
- Required revalidation includes exact hashes and 50-feature order, calibrated V3 parity, issue-time line
- The merge is worth it only as staged selective adaptation after Repository B's immediate P0 blockers are

FINAL_ONE_PARAGRAPH_RECOMMENDATION: Do not submit either repository unchanged and do not merge for the sal